Orchestrator’s reference

Problem - A damaged embedded CNC controller was discovered from an abandoned research facility. The machine was mid-job when the power got cut. The engineers said the machine was engraving something important before it died. Can you recover what it was making and find the flag in the process? The flag contains characters f' to identify when you see it. The only file the team was able to recover from the CNC machine was the binary file last loaded onto the embedded controller, provided to you. Good Luck!

IDEAL PLAYER STEPS:

The player is given the .bin file

The linux tool strings reveals the printable content from the .bin file

```
lucfrbrkr4@Ares: ~/Downloads/GCODE/Orchestrator
AXIOM-CNC fw v2.3.1
AXIOM-EMB-32 (c) 2021 Axiom Precision Ltd
job_buffer_load: segment %d of %d
WARNING: CRC mismatch on segment header
FAULT: watchdog timeout during job_exec at flash+0x00001000
job_buffer: packet format [4B:length][1B:seg_id][NB:data] x4 segments
spindle_on: PWM duty %d%%
axis_limit_exceeded: halting
SAFE MODE ACTIVE
JOB BUFFER FRAGMENTED
[AXIOM DEBUG] config struct OK (sizeof=34 bytes, magic=0xAA104D43): magic=0x%08X
 baud=%d x_max=%.1f y_max=%.1f z_max=%.1f feed_max=%d spindle_max=%d cal_reserve
d=%s
cal_reserved (0x0C18): DO NOT MODIFY
JBUFHDR5SEG4
 n;^
Qkkbal
i]Wb
9a&g
MGiI
wn>Jj
#.zf
+o*7
AXIOM_END
```

The player notices "AXIOM-CNC" and understands that this is an operational binary file for the controller. There are some errors that have come up during the running of the binary file:

- A CRC mismatch
- A watchdog timeout during execution at flash address 0x00001000
- A halt log
- A packet format of 4B length, 1B seg_id, NB data
- Safe mode activation and Job Buffer Fragmented
- An interesting string JBUFHDR5SEG4

The CRC mismatch is a fake error raise. Inspection of the same in the hex format of the bin file shows no errors.

The player associates the timeout occurring at the flash address. And this was when the power was cut.

Next some config's addresses are provided, given starting address and magic number. Also the player notices that the last 8 bytes of the allocated 34 bytes is not to meddled with

HEXDUMP THE BIN FILE:

The player then proceeds to perform hexdump of the binary file, which gives a rather structured hexdump

Command : `hexdump -C controller_fw.bin >> hex_dump.txt`

```
lucfrbrkr4@Ares: ~/Downloads/GCODE/Orchestrator
00000990  34 20 62 79 74 65 73 2c  20 6d 61 67 69 63 3d 30  |4 bytes, magic=0|
000009a0  78 41 41 31 30 34 44 34  33 29 3a 20 6d 61 67 69  |xAA104D43): magi|
000009b0  63 3d 30 78 25 30 38 58  20 62 61 75 64 3d 25 64  |c=0x%08X baud=%d|
000009c0  20 78 5f 6d 61 78 3d 25  2e 31 66 20 79 5f 6d 61  | x_max=%.1f y_ma|
000009d0  78 3d 25 2e 31 66 20 7a  5f 6d 61 78 3d 25 2e 31  |x=%.1f z_max=%.1|
000009e0  66 20 66 65 65 64 5f 6d  61 78 3d 25 64 20 73 70  |f feed_max=%d sp|
000009f0  69 6e 64 6c 65 5f 6d 61  78 3d 25 64 20 63 61 6c  |indle_max=%d cal|
00000a00  5f 72 65 73 65 72 76 65  64 3d 25 73 00 63 61 6c  |_reserved=%s.cal|
00000a10  5f 72 65 73 65 72 76 65  64 20 28 30 78 30 43 31  |_reserved (0x0C1|
00000a20  38 29 3a 20 44 4f 20 4e  4f 54 20 4d 4f 44 49 46  |8): DO NOT MODIF|
00000a30  59 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |Y...............|
00000a40  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |................|
*
00000c00  43 4d 10 aa 00 c2 01 00  00 00 48 43 00 00 48 43  |CM........HC..HC|
00000c10  00 00 48 42 90 01 c0 5d  f1 4c 3b a7 2e 91 c4 08  |..HB...].L;.....|
00000c20  04 de 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |................|
00000c30  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |................|
*
00001000  4a 42 55 46 48 44 52 35  53 45 47 34 92 0f 00 00  |JBUFHDR5SEG4....|
00001010  03 d9 3f 5e c0 14 a5 eb  3c d8 46 13 c4 41 ff b0  |..?^....<.F..A..|
00001020  61 9f 39 5e 9d 0e d0 9c  41 be 01 16 e3 6b d3 91  |a.9^....A....k..|
00001030  4f d1 3c 54 d5 5a b1 f0  3c c5 78 12 ad 69 a1 f4  |O.<T.Z..<.x..i..|
00001040  28 ab 79 15 97 1e a1 f4  38 c1 46 7c 97 1e b1 9c  |(.y.....8.F|....|
00001050  39 c4 75 15 95 17 a8 f2  3d c0 6c 62 96 1d a7 ea  |9.u.....=.lb....|
```

They search for the magic bytes `43 4D 10 AA` - the little-endian representation of `0xAA104D43` from the format string. They find them at `0x0C00`, confirming this is the config struct. They then jump directly to `0x0C18` as instructed by the warning string and read 8 bytes:

`f1 4c 3b a7 2e 91 c4 08`

These are non-printable and never appeared in the `strings` output. The player recognises this as the `cal_reserved` field - the XOR key. They also note the surrounding fields confirm the struct is legitimate, readable baud rate value, plausible axis limits as floats, clean structure throughout.

Then the player sees the string from the string dump JBUFHDR5SEG4, and understands that it's the Job Buffer's image stored in the non-volatile memory instead of the volatile memory, but then understands this is an industrial CNC machine with such built in safety features as as said previously in the header,

```
SAFE MODE ACTIVE
JOB BUFFER FRAGMENTED
```

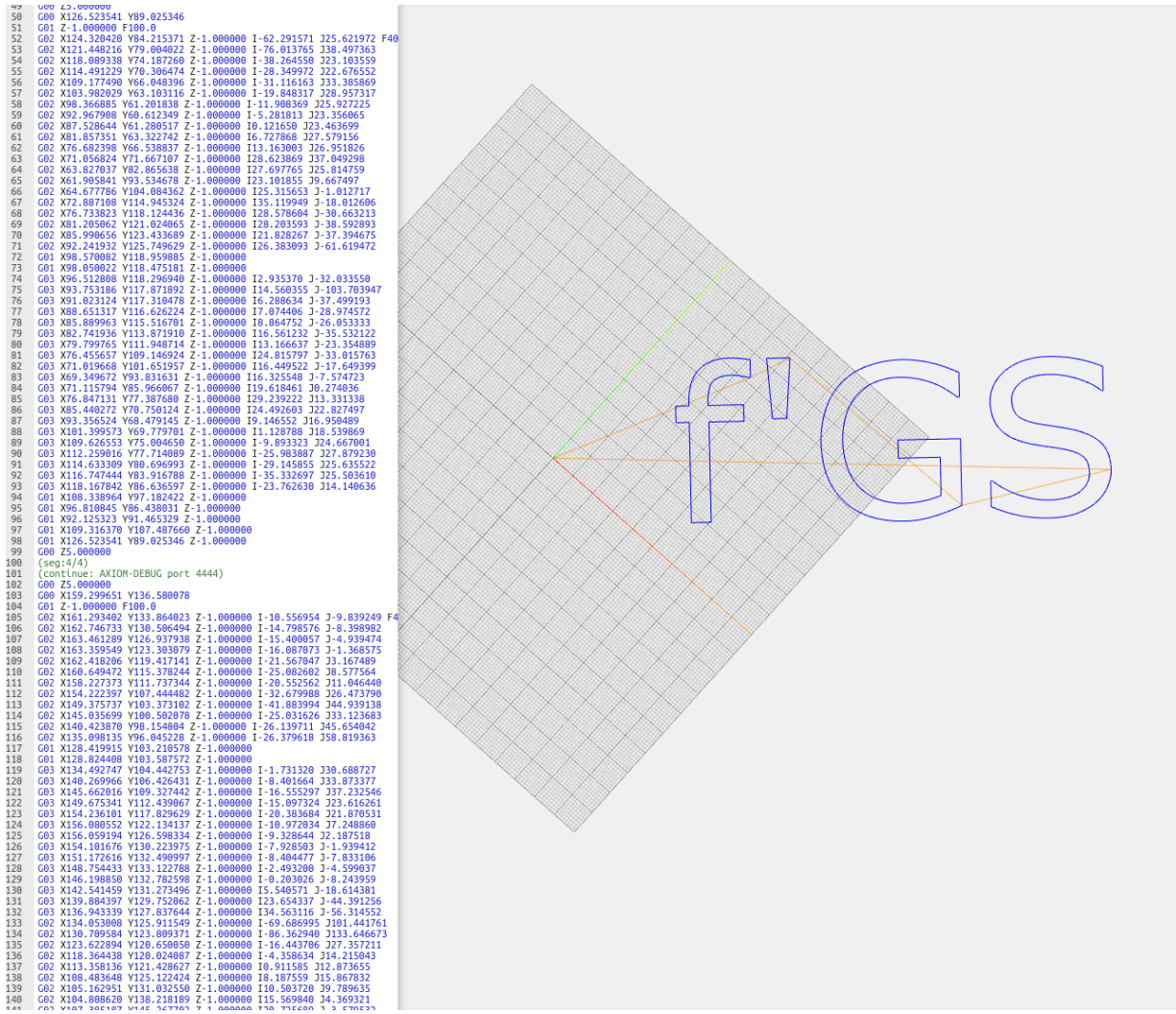
understanding that there are 4 segments of program code (GCODE) each in a length-prefix format with a header of 5 bytes of 4 bytes saying its length and 1 byte saying its segment_id.

DISCOVERING THE GCODE:

The player now has a key and a location. They go to `0x1000` - where the watchdog fault pointed, skip 12 bytes of the Job Buffer (JBUF) and see structured binary data. The first 4 bytes `92 0F 00 00` read as little-endian integer `3986` - a plausible data length. The 5th byte `03` is a segment ID. Then a long run of garbage bytes follows. The pattern repeats. The player recognises the `[length][id][data]` packet structure. Combined with `segment %d of %d` from the strings output and `seg_count = 4` visible in the config struct, they understand there are 4 XOR-encoded GCode segments stored in shuffled order.

VISUALIZING THE GCODE PLOT:

The player then uses an open source software or web app to plot what the GCODE was trying to do. The plot reveals f’GS and is the whole FLAG.



```
G00 Z5.000000
G00 X126.523541 Y89.025346
G01 Z-1.000000 F100.0
G02 X124.320420 Y84.215371 Z-1.000000 I-62.291571 J25.621972 F40
G02 X121.448216 Y79.004022 Z-1.000000 I-76.013765 J38.497363
G02 X118.089338 Y74.187260 Z-1.000000 I-38.264550 J23.103559
G02 X114.491229 Y70.306474 Z-1.000000 I-28.349972 J22.676552
G02 X109.177490 Y66.048396 Z-1.000000 I-31.116163 J33.385869
G02 X103.982029 Y63.103116 Z-1.000000 I-19.848317 J28.957317
G02 X98.366885 Y61.201838 Z-1.000000 I-11.908369 J25.927225
G02 X92.967908 Y60.612349 Z-1.000000 I-5.281813 J23.356065
G02 X87.528644 Y61.280517 Z-1.000000 I0.121650 J23.463699
G02 X81.857351 Y63.322742 Z-1.000000 I6.727868 J27.579156
G02 X76.682398 Y66.538837 Z-1.000000 I13.163003 J26.951826
G02 X71.056824 Y71.667107 Z-1.000000 I28.623869 J37.049298
G02 X63.827037 Y82.865638 Z-1.000000 I27.697765 J25.814759
G02 X61.905841 Y93.534678 Z-1.000000 I23.101855 J9.667497
G02 X64.677786 Y104.084362 Z-1.000000 I25.315653 J-1.012717
G02 X72.887108 Y114.945324 Z-1.000000 I35.119949 J-18.012606
G02 X76.733823 Y118.124436 Z-1.000000 I28.578604 J-30.663213
G02 X81.205062 Y121.024065 Z-1.000000 I28.203593 J-38.592893
G02 X85.990656 Y123.433689 Z-1.000000 I21.828267 J-37.394675
G02 X92.241932 Y125.749629 Z-1.000000 I26.383093 J-61.619472
G01 X98.570082 Y118.959885 Z-1.000000
G01 X98.050022 Y118.475181 Z-1.000000
G03 X96.512808 Y118.296940 Z-1.000000 I2.935370 J-32.033550
G03 X93.753186 Y117.871892 Z-1.000000 I14.560355 J-103.703947
G03 X91.023124 Y117.310478 Z-1.000000 I6.288634 J-37.499193
G03 X88.651317 Y116.626224 Z-1.000000 I7.074406 J-28.974572
G03 X85.889963 Y115.516701 Z-1.000000 I8.864752 J-26.053333
G03 X82.741936 Y113.871910 Z-1.000000 I16.561232 J-35.532122
G03 X79.799765 Y111.948714 Z-1.000000 I13.166637 J-23.354889
G03 X76.455657 Y109.146924 Z-1.000000 I24.815797 J-33.015763
G03 X71.019668 Y101.651957 Z-1.000000 I16.449522 J-17.649399
G03 X69.349672 Y93.831631 Z-1.000000 I16.325548 J-7.574723
G03 X71.115794 Y85.966067 Z-1.000000 I19.618461 J0.274036
G03 X76.847131 Y77.387680 Z-1.000000 I29.239222 J13.331338
G03 X85.440272 Y70.750124 Z-1.000000 I24.492603 J22.827497
G03 X93.356524 Y68.479145 Z-1.000000 I9.146552 J16.958489
G03 X101.399573 Y69.779701 Z-1.000000 I1.128788 J18.539869
G03 X109.626553 Y75.004650 Z-1.000000 I-9.893323 J24.667001
G03 X112.259016 Y77.714089 Z-1.000000 I-25.983887 J27.879230
G03 X114.633309 Y80.696993 Z-1.000000 I-29.145855 J25.635522
G03 X116.747444 Y83.916788 Z-1.000000 I-35.332697 J25.503610
G03 X118.167842 Y86.636597 Z-1.000000 I-23.762630 J14.140636
G01 X108.338964 Y97.182422 Z-1.000000
G01 X96.810845 Y86.438031 Z-1.000000
G01 X92.125323 Y91.465329 Z-1.000000
G01 X109.316370 Y107.487660 Z-1.000000
G01 X126.523541 Y89.025346 Z-1.000000
G00 Z5.000000
(seg:4/4)
(continue: AXIOM-DEBUG port 4444)
G00 Z5.000000
G00 X159.299651 Y136.580078
G01 Z-1.000000 F100.0
G02 X161.293402 Y133.864023 Z-1.000000 I-10.556954 J-9.839249 F4
G02 X162.746733 Y130.506494 Z-1.000000 I-14.798576 J-8.398982
G02 X163.461289 Y126.937938 Z-1.000000 I-15.400057 J-4.939474
G02 X163.359549 Y123.303079 Z-1.000000 I-16.087073 J-1.368575
G02 X162.418206 Y119.417141 Z-1.000000 I-21.567047 J3.167489
G02 X160.649472 Y115.378244 Z-1.000000 I-25.082602 J8.577564
G02 X158.227373 Y111.737344 Z-1.000000 I-20.552562 J11.046440
G02 X154.222397 Y107.444482 Z-1.000000 I-32.679988 J26.473790
G02 X149.375737 Y103.373102 Z-1.000000 I-41.883994 J44.939138
G02 X145.035699 Y100.502078 Z-1.000000 I-25.031626 J33.123683
G02 X140.423870 Y98.154804 Z-1.000000 I-26.139711 J45.654042
G02 X135.098135 Y96.045228 Z-1.000000 I-26.379618 J58.819363
G01 X128.419915 Y103.210578 Z-1.000000
G01 X128.824408 Y103.587572 Z-1.000000
G03 X134.492747 Y104.442753 Z-1.000000 I-1.731320 J30.688727
G03 X140.269966 Y106.426431 Z-1.000000 I-8.401664 J33.873377
G03 X145.662016 Y109.327442 Z-1.000000 I-16.555297 J37.232546
G03 X149.675341 Y112.439067 Z-1.000000 I-15.097324 J23.616261
G03 X154.236101 Y117.829629 Z-1.000000 I-20.383684 J21.870531
G03 X156.080552 Y122.134137 Z-1.000000 I-10.972034 J7.248860
G03 X156.059194 Y126.598334 Z-1.000000 I-9.328644 J2.187518
G03 X154.101676 Y130.223975 Z-1.000000 I-7.928503 J-1.939412
G03 X151.172616 Y132.490997 Z-1.000000 I-8.404477 J-7.833106
G03 X148.754433 Y133.122788 Z-1.000000 I-2.493200 J-4.599037
G03 X146.198850 Y132.782598 Z-1.000000 I-0.203026 J-8.243959
G03 X142.541459 Y131.273496 Z-1.000000 I5.540571 J-18.614381
G03 X139.884397 Y129.752062 Z-1.000000 I23.654337 J-44.391256
G03 X136.943339 Y127.837644 Z-1.000000 I34.563116 J-56.314552
G02 X134.053008 Y125.911549 Z-1.000000 I-69.686995 J101.441761
G02 X130.709584 Y123.809371 Z-1.000000 I-86.362940 J133.646673
G02 X123.622894 Y120.650050 Z-1.000000 I-16.443706 J27.357211
G02 X118.364438 Y120.024087 Z-1.000000 I-4.358634 J14.215043
G02 X113.358136 Y121.428627 Z-1.000000 I0.911585 J12.873655
G02 X108.483648 Y125.122424 Z-1.000000 I8.187559 J15.867832
G02 X105.162951 Y131.032550 Z-1.000000 I10.503720 J9.789635
G02 X104.808620 Y138.218189 Z-1.000000 I15.569840 J4.369321
```

``` That’s it ```

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